

GRAND FINALE - PROBLEM STATEMENT

CRITICAL SYSTEMS & REAL-TIME OPERATIONS

AIRPORT OPERATIONS CONTROL CENTER

1. Challenge Brief

Build a real-time Airport Operations Control Center that transforms the provided multi-table airport dataset into a unified, interactive operational interface for monitoring flights, gates, baggage, passengers, security, maintenance, staff and other airport activities.

2. Background

Modern airports operate as complex, time-sensitive ecosystems in which multiple departments must coordinate simultaneously. Operational teams need a single interface that can convert fragmented data into clear situational awareness, highlight disruptions, and support rapid operational decisions.

3. Your Mission

Design and develop a production-quality frontend that behaves like an Airport Operations Control Center. The application should unify the supplied datasets, expose meaningful relationships between them, and simulate a live operational environment rather than presenting a collection of static charts.

4. Provided Dataset

Participants will work with the provided multi-table airport dataset, including:

- **flights.csv** - flight schedules, status and operational flight information.
- **gate_events.csv** - gate activity and gate-related events.
- **baggage.csv** - baggage handling and movement information.
- **passengers.csv** - passenger-related operational data.
- **security_screening.csv** - security screening activity.
- **maintenance_logs.csv** - maintenance events and operational logs.
- **staff_shifts.csv** - staff scheduling and shift information.
- **retail_transactions.csv** - airport retail transaction activity.
- **data_dictionary.md** - field definitions and dataset reference.

5. Core Product Expectations

- Unified operations dashboard with clear airport-wide status visibility.
- Flight monitoring with search, filters, status indicators and detailed views.
- Gate and operational resource visibility, including conflicts or abnormal conditions where supported by the data.

- Operational views for baggage, passengers, security, maintenance and staff.
- Meaningful cross-table relationships instead of isolated dataset pages.
- Alerts, exceptions or incident-style indicators that help users identify what requires attention.
- Simulated real-time behaviour using the supplied static dataset, such as timed status changes, event feeds or operational updates.
- Responsive, polished and information-dense UI suitable for a mission-critical operations environment.
- Useful interactions such as filtering, sorting, drill-down views, timelines, detail panels and operational controls.

6. Functional Standard

A visually attractive dashboard alone is not sufficient. The final solution must demonstrate meaningful functionality and operational thinking. Participants should decide how the supplied data can be transformed into a coherent workflow for airport operators and how users can move from high-level monitoring to detailed investigation.

7. Real-Time Simulation

The provided dataset may be static, but the application must create the experience of a changing operational environment. Teams may simulate event progression, status changes, alerts, queue updates, timelines or other live behaviours using frontend logic. The simulation should remain consistent with the supplied data and the product experience.

8. Submission Deliverables

- Working deployed application link.
- Complete GitHub repository containing the source code.
- Clear README with setup instructions, technology stack and feature summary.
- Application must use the provided dataset as the primary challenge data source.
- Final solution must be ready for live demonstration and judge interaction.

9. Evaluation Focus

Area	What Judges Will Look For
Operational Thinking	How effectively the interface represents real airport operations and connects multiple datasets.
Functionality	Depth and usefulness of interactions, monitoring, drill-downs, alerts and controls.
Data Integration	Quality of relationships and insights created across the provided tables.
UI/UX	Clarity, hierarchy, usability and suitability for a high-pressure control-center environment.
Real-Time Experience	Credibility and usefulness of the simulated live operational behaviour.
Code Quality	Structure, maintainability and implementation quality.